

LITERATURE STATUS FOR ARXIV:2206.03794

AGENT_LANE_10

SOURCE QUESTION

At the end of the Introduction of arXiv:2206.03794, Kakol–Marciszewski–Sobota–Zdomskyy ask whether

$$C(\beta\mathbb{N} \times \beta\mathbb{N})$$

has a complemented infinite-dimensional subspace not isomorphic to any of the following five spaces:

$$C(\beta\mathbb{N} \times \beta\mathbb{N}), \quad C(\beta\mathbb{N}), \quad c_0, \quad c_0 \oplus C(\beta\mathbb{N}), \quad c_0(C(\beta\mathbb{N})).$$

ANSWER

Plebanek–Rondos–Sobota, arXiv:2405.19120, explicitly revisit this Alspach–Galego/Candido problem. Their Corollary 1.4 states that

$$C(\beta\omega \times \beta\omega)$$

contains a complemented subspace isomorphic to $C([0, 1]^\kappa)$ for every cardinal $1 \leq \kappa \leq \mathfrak{c}$. Taking $\kappa = 1$, it contains a complemented copy of $C[0, 1]$.

The space $C[0, 1]$ is not among the five listed isomorphism classes. Four of the listed spaces are nonseparable because they contain a copy of $C(\beta\mathbb{N}) \cong \ell_\infty$, while $C[0, 1]$ is separable. The remaining listed space is c_0 , and $C[0, 1] \not\cong c_0$ because $c_0^* = \ell_1$ is separable whereas $C[0, 1]^*$, the space of finite signed regular Borel measures on $[0, 1]$, is nonseparable.

SCOPE

Thus arXiv:2405.19120 gives a positive answer to the existence question in arXiv:2206.03794. This does not classify all complemented subspaces of $C(\beta\omega \times \beta\omega)$.

REFERENCES

- [1] J. Kakol, W. Marciszewski, D. Sobota, and L. Zdomskyy, *On complementability of c_0 in spaces $C(K \times L)$* , arXiv:2206.03794.
- [2] G. Plebanek, J. Rondos, and D. Sobota, *Complemented subspaces of Banach spaces $C(K \times L)$* , arXiv:2405.19120.